

Reductions

Independent Set - Vertex Cover

- Take the complement of the graph of one's input and give to the other

VERTEX COVER and INDEPENDENT SET

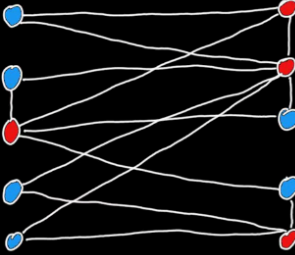
Goal: Show that $\text{VERTEX COVER} \equiv_P \text{INDEPENDENT SET}$

Definition (VERTEX COVER)
Given a graph $G=(V, E)$ and integer K , is there a subset of vertices $S \subseteq V$ such that $|S| \leq k$ and for each edge at least one of its endpoints is in S .

• a vertex cover of size 4
there is no vertex cover of size 3
For $k=4$, answer would be "yes"
For $k=3$, answer would be "no"

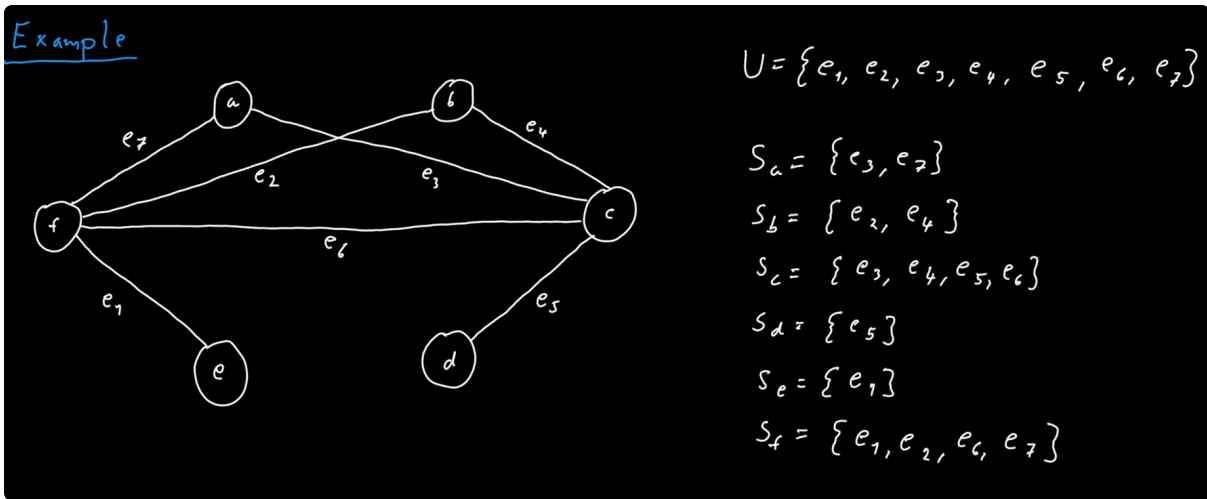
Definition (INDEPENDENT SET)
Given a graph $G=(V, E)$ and integer K , is there a subset of vertices $S \subseteq V$ such that $|S| \geq k$ and for each edge at most one of its endpoints is in S .

• an independent set of size 6
there is no independent set of size 7
For $k=6$, answer would be "yes"
For $k=7$, answer would be "no"



Set Cover - Vertex Cover

- $1 \rightarrow 2 \Rightarrow$ construct a graph and find the vertex cover
- $2 \rightarrow 1 \Rightarrow$ construct the sets from the graph and find the set cover



3 SAT to Independent Set

- In each clause, there will be only 3 variables
- draw graphs from the input
 - for each clause, draw a triangle graph
 - from each triangle, connect x to other triangles if x'
 - i.e., if there is x in one triangle, draw an edge to all x' in other triangles
- After this, find the independent set of size k in this
 - In each triangle, we can choose only one vertex (it is connected to the other 2 in the triangle)
 - Since, we connected all pairs of x and x' , it ensures that x and x' will

never be true together

- For the existence of an independent set of size k in the graph, there should be one vertex in each triangle $\Rightarrow$ One true in each clause

Satisfiability

Claim: $3SAT \leq_P INDEPENDENT SET$

Given an instance ϕ of $3SAT$, construct an instance $\langle G, K \rangle$ of $INDEPENDENT SET$ such that G has an independent set of size at least K iff ϕ is satisfiable.

Construction

$\phi = (\bar{x}_1 \vee x_2 \vee x_3) \wedge (x_1 \vee \bar{x}_2 \vee x_3) \wedge (\bar{x}_1 \vee x_2 \vee x_4)$

G :

$K=3$

1. Create 3 vertices for each clause
2. Connect these 3 vertices to a triangle
3. Connect every literal to each of its negations
4. Set K equal to number of clauses

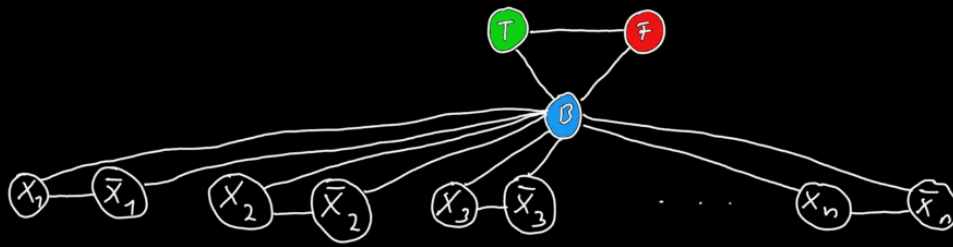
Proof

$\Rightarrow$ " Let S be independent set of size K
 S must contain a vertex from each triangle
Set corresponding literals to true
truth assignment is consistent and satisfies ϕ

$\Leftarrow$ " Given a satisfier

3 SAT - 3 Colorability

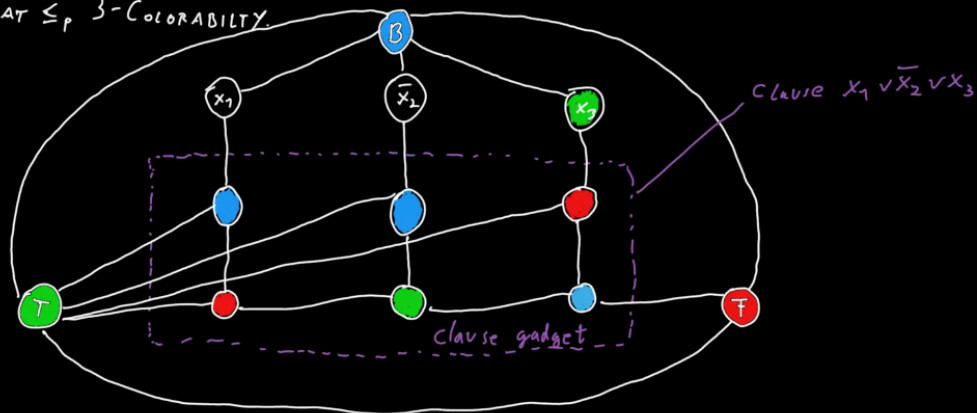
Theorem: $3SAT \leq_p 3-COLORABILITY$.



3-COLORABILITY

Given a graph G . Is it possible to color all vertices of G using only three colors such that no two adjacent vertices have the same color.

Theorem: $3SAT \leq_p 3-COLORABILITY$.



If the CNF formula has a satisfying assignment, we can color vertices representing literals accordingly. Can also extend the coloring to all gadgets.

If we have a valid coloring, the colors of vertices representing literals imply a satisfying assignment for the CNF formula.

SAT to 3 SAT

- for clauses of form $(l) \rightarrow (l \text{ or } l \text{ or } l)$
- for clauses of form $(l1 \text{ or } l2) \rightarrow (l1 \text{ or } l2 \text{ or } l2)$
- for clauses with length, $k > 3$
 - add extra $k - 3$ variables: $y1, y2, \dots, y(k-3)$
 - $(l1 \text{ or } l2 \text{ or } l3 \text{ or } \dots \text{ or } ln)$

- update the clauses as
 - $(l_1 \vee l_2 \vee y_1)$
 - $(\neg y_1 \vee l_3 \vee y_2)$
 - $(\neg y_2 \vee l_4 \vee y_3)$
 - ...
 - $(\neg y_{k-3} \vee l_{k-1} \vee l_k)$
 - and take the ANDs of them
- If any of the variable in a clause is true, we will make both the Ys in that clause as false resulting in making all other clauses true
- If all other variables in a clause is false, we will make the y in the clause as true, making the whole input true only if any of the other clause is true

3 SAT to SAT

- 3 SAT is a special case of SAT

SAT - Clique

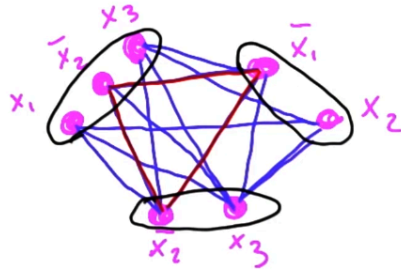
- Construct a graph

- For each literal in each clause, add a vertex
- Add edges from vertices of one clause to another such that
 - do not add vertex from x to x'
 - add all other possible edges
- In this graph, inside one clause, no vertices are connected
- And between clauses, all pairs other than (x, x') pairs are connected
- If the CNF is satisfiable, then there will be a clique of size k , because
 - We all edges other than x, x' pair and edges within the clause
 - this means, for a clique of size k to exist, each vertex should be taken from each clause. And since there is no (x, x') pair, these chosen vertices

are consistent

$$\underbrace{(x_1 \vee \bar{x}_2 \vee x_3)}_{\text{clause 1}} \wedge \underbrace{(\bar{x}_1 \vee x_2)}_{\text{clause 2}} \wedge \underbrace{(\bar{x}_2 \vee x_3)}_{\text{clause 3}}$$

m clauses



QUIZ: What if the graph has a clique of size m ?

- ☐ Contains at least one vertex from each "clause group"
- ☐ Clique can contain multiple vertices from a "clause group"

3 SAT to Vertex Cover